Relative Susceptibility of Six Soybean Caterpillars[1] to a Standard Preparation of *Bacillus thuringiensis* var. *kurstaki*[2]

C. M. Ignoffo, D. L. Hostetter, R. E. Pinnell, and C. Garcia

Biological Control of Insects Research Unit, North Central Region, Agricultural Research Service, USDA, Columbia, MO 65201

ABSTRACT

The LC_{50} of a standard preparation of *B. thuringiensis* Berliner var. *kurstaki* against 1-day-old larvae of *Trichoplusia ni* (Hübner) was 90 ng/cm² of diet surface. One-day-old larvae of *Anticarsia gemmatalis* Hübner and *Plathypena scabra* (F.) were >10 × more susceptible than *T. ni*; *Pseudoplusia includens* (Walker) was equally susceptible, and *Spodoptera exigua* (Hübner) and *Heliothis zea* (Boddie) were ca. 3 × less susceptible. The LC_{50} against 4-day-old *T. ni* was 1800 ng/cm² of diet surface, and the relative ranking for susceptibility of 4-day-old larvae of the other species was similar to that for 1-day-old larvae. Also ranking for susceptibility based on weight of larvae that survived was identical to that based on larval mortality.

There are several lepidopteran pests or potential pests of soybeans in the United States. In the Southeast and Mid-south, the major defoliating caterpillars are the velvetbean caterpillar, *Anticarsia gemmatalis* Hübner; the soybean looper, *Pseudoplusia includens* (Walker), and occasionally the beet armyworm, *Spodoptera exigua* (Hübner). In the Midwest, the major defoliator is the green cloverworm, *Plathypena scabra* (F.), though species of Plusiinae such as the cabbage looper, *Trichoplusia ni* (Hübner), also occasionally damage Midwest soybeans. The major pod-feeding pests found throughout the soybean growing areas are the bollworm, *Heliothis zea* (Boddie), and the tobacco budworm, *H. virescens* (F.). If a safe, selective microbial insecticide can be used effectively against these pests, we may avoid, or at least minimize, the kinds of problems that have developed in other ecosystems, notably cotton (van den Bosch 1972), where broad-spectrum chemical insecticides have been used against some of these same species.

Our preliminary laboratory tests (unpublished 1971), evidence from published articles (Rogoff et al. 1969, Sommerville et al. 1970, Libby and Chapman 1971, Dulmage and Rhodes 1971), and label registrations indicate that some of the cited species are susceptible to the microbial insecticide *Bacillus thuringiensis* Berliner. However, information was not available concerning the susceptibility of *P. scabra, P. includens*, and *A. gemmatalis*; and no information was available about the relative susceptibility of any of the species to *B. thuringiensis*.

Formulations of *B. thuringiensis* are currently registered for controlling *T. ni* on many truck and row crops. Consequently, we evaluated the relative susceptibility of the soybean caterpillar pests to *B. thuringiensis* using *T. ni* as our standard. If the susceptibility of these soybean caterpillars was equivalent to that of *T. ni*, then we might anticipate levels of field control equivalent to that routinely observed for *T. ni* on row crops.

Methods and Materials.—*Bacillus thuringiensis Standard*.—The standard preparation of *B. thuringiensis* var. *kurstaki* we used in all our tests was a wettable powder provided by Dr. H. T. Dulmage, Brownsville, Texas (11/25/71). This preparation, designated HD-1-S-1971, contained 25,000 spores/μg and 16.0 IU/μg when it was assayed against 4-day-old *T. ni* larvae (Dulmage et al. 1971). The isolate, originally described as var. *alesti* (Dulmage 1970), was later identified as var. *kurstaki* (Dulmage and Rhodes 1971). The standard preparation was thoroughly ground (teflon-pestle tissue grinder) before it was suspended in phosphate-buffered saline solution (NaCl 0.85%, K_2HPO_4 0.6%, KH_2PO_4 0.3%, Tween-80 0.025%). All subsequent dilutions for bioassay were made in buffered solution.

Bioassay Technique.—Larvae of all species were exposed to a semisynthetic diet that was untreated or had the surface treated with *B. thuringiensis* as previously described (Ignoffo 1966, Ignoffo et al. 1968). The appropriate dose was applied by using a Centaur™ micropipette to dispense 100 μl of the suspension on 7.7 cm² of diet surface (Ignoffo and Boening 1970). Then this drop was evenly distributed over the diet surface with a sterile glass rod, and the surface was air-dried. One larva each was confined to each cubicle of a 50-cubicle plastic tray, and the trays were incubated at 30±1°C. Mortality was recorded after 7 days. Surviving larvae from each replicate were pooled and weighed when they were 11 days old.

Test Insects.—One-, and 4-day-old larvae of the following pest species of soybean caterpillars were used in our studies: *A. gemmatalis, H. zea, P. scabra, P. includens, S. exigua*, and *T. ni*. Larvae of *T. ni* were our standard for comparing levels of relative susceptibility. All species except *P. scabra* and *A. gemmatalis* were transferred from untreated to treated diets after 24 or 96 h of feeding and thereafter continuously exposed on diets surface-treated with *B. thuringiensis*. Larvae of *A. gemmatalis* and *P. scabra* were fed on soybean leaflets prior to bioassay since initial feeding on natural food was required for subsequent survival

[1] Lepidoptera: Noctuidae.

[2] This paper reports the results of research only. Mention of a pesticide does not constitute a recommendation for use by the USDA nor does it imply registration under FIFRA as amended. Also mention of a proprietary product does not constitute an endorsement by the USDA. Received for publication June 7, 1976.

on our semisynthetic diet. A basic wheat germ diet was used to rear and test *H. zea*, *T. ni*, *P. includens*, and *S. exigua* (Ignoffo 1966). Alfalfa was added to this diet for *T. ni* (Ignoffo 1966), and soy protein was added for rearing *A. gemmatalis* and *P. scabra* (Greene et al. 1976, in press).

The number of 1-, or 4-day-old larvae and replicates, in parentheses, for evaluating relative susceptibility was as follows:

	1-day-old		4-day-old	
	Untreated	Treated	Untreated	Treated
T. ni	544 (11)	516 (11)	250 (5)	246 (5)
H. zea	150 (3)	145 (3)	149 (3)	123 (3)
S. exigua	132 (3)	140 (3)	150 (3)	149 (3)
A. gemmatalis	185 (4)	189 (4)	149 (3)	150 (3)
P. includens	144 (3)	138 (3)	150 (3)	145 (3)
P. scabra	183 (4)	198 (4)	143 (4)	148 (4)

Generally, 3–4 replicates consisting of 50 larvae/replicate were used for all species except where specifically noted. More replicates were used for *T. ni* because this was the standard insect, and it was included each time a bioassay was conducted. The LC_{50} values were interpolated from graphed points. Slope of the dose-mortality line for *T. ni* was based on at least 4 replicates of ca. 5 points/line. The slope of the line for all other species was based on at least 6 points falling between 20 and 80% mortality.

RESULTS.—*Susceptibility of T. ni*.—The estimated LC_{50} for 1-, or 4-day-old *T. ni* larvae continuously exposed to diets surface-treated with *B. thuringiensis* var. *kurstaki* was 90 ng/cm² or 1800 ng/cm², respectively (Table 1). Thus 1-day-old, 1st-instars were ca. 20× more susceptible than 4-day-old, 2nd-instars. On a weight basis, however, the estimated LC_{50} for 1-day-old *T. ni* larvae with an average initial weight of 0.5±0.1 mg was 182 ng/cm² and that for 4-day-old larvae with an average initial weight 10.0±1.5 mg was 180 ng/cm².

Relative Susceptibility of Soybean Caterpillars.—One-day-old and 4-day-old larvae of *P. scabra* and *A. gemmatalis* were more susceptible to *B. thuringiensis* var. *kurstaki* than *T. ni* (Tables 2 and 3). Larvae of *P. includens* were equal in susceptibility, and *H. zea* and *S. exigua* were less susceptible than *T. ni*. The most and least sensitive species were *P. scabra* (ca. 100 × more sensitive) and *H. zea* (ca. 3–4 less sensitive), respectively. The range in susceptibility for both 1-, and 4-day-old larvae was thus >300×.

Generally, the magnitudes of susceptibility relative to *T. ni* were similar for both 1-, and 4-day-old larvae of the same species except for *A. gemmatalis*, e.g., the relative susceptibility of 1-, or 4-day-old *H. zea* to *T. ni* was 3.18 and 3.40, respectively, a difference of 6.5%. Differences for *P. scabra, S. exigua*, and *P. includens* were 0.0, 5.8, and 15.6%, respectively. In contrast, there was ca. a 9-fold difference in relative susceptibility between 1-, and 4-day-old larvae of *A. gemmatalis*. This may indicate that *A. gemmatalis* larvae have increasing resistance to *B. thuringiensis* as they mature.

Weight of Survivors.—All larvae, whether exposed as 1-, or 4-day-old larvae to diets treated with *B. thuringiensis*, weighed considerably less than their counterparts exposed on untreated diets (Table 4). Reductions in weight gains are usual when larvae are exposed continuously to diets treated with *B. thuringiensis* (Ignoffo et al. 1968). Differences in weight between treated and untreated 1-day-old larvae ranged from ca. 3-fold (*S. exigua*) to 22-fold (*P. scabra*). Differences in weight of 4-day-old larvae ranged from none (*S. exigua*) to ca. 27-fold (*P. inclu-*

Table 1.—Average percentage mortality of 1-, and 4-day-old *T. ni* larvae continuously exposed to diets surface-treated with *Bacillus thuringiensis* var. *kurstaki*.[a]

Rate[b] ng/cm²	No. larvae	Avg % mortality ± $S\bar{X}$
	1-day-old larvae[c]	
0.0	199	4.2±0.8
39.0	192	25.5±8.3
64.9	196	41.5±8.1
90.9	194	50.5±11.2
129.9	190	70.0±8.9
	4-day-old larvae[d]	
0.0	450	2.0±0.9
649.4	400	25.2±7.4
1298.7	300	44.7±10.9
3246.8	300	74.3±8.3

[a] Standard preparation of *Bacillus thuringiensis* var. *kurstaki* HD-1-S-1971 contained 16 IU/μg and ~25,000 viable spores/μg.
[b] Actual weight of *B. thuringiensis* dispensed in volume of 0.1 ml/7.7 cm² of diet surface.
[c] Four replicates, ca. 50 larvae/replicate; initial weight, 0.5±0.1 mg/larva.
[d] Six-9 replicates; ca. 50 larvae/replicate; initial weight, 10.0±1.5 mg/larva.

Table 2.—Average percentage mortality (± $S\bar{X}$), as well as estimated LC_{50}, and relative susceptibility of 1-day-old larvae continuously exposed to untreated diets or diets surface-treated with *Bacillus thuringiensis* var. *kurstaki*.

	% mortality		Estimated	
Species	Untreated larvae	Treated[a] larvae	LC_{50} (ng/cm²)	Relative susceptibility[b]
Trichoplusia ni	3.2±0.8	63.5±5.6	85.7	1.00
Heliothis zea	1.4±0.6	20.7±4.9	272.7	3.18
Spodoptera exigua	3.4±0.3	25.3±10.4	233.8	2.73
Pseudoplusia includens	6.3±2.2	40.9±4.3	93.5	1.09
Anticarsia gemmatalis	1.2±1.0	76.9±6.4[c]	7.8	0.09
Plathypena scabra	2.7±3.1	41.0±11.8[d]	0.9	0.01

[a] Larvae exposed to 90.9 ng/cm² of diet except where noted.
[b] Based on LC_{50} of *T. ni*.
[c] Larvae exposed to 11.7 ng/cm² of diet.
[d] Larvae exposed to 0.8 ng/cm².

Table 3.—Average percentage mortality (± S$\bar{X}$), estimated LC_{50}, and relative susceptibility of 4-day-old larvae continuously exposed to untreated diets or diets surface-treated with *Bacillus thuringiensis* var. *kurstaki*.

	% mortality		Estimated	
Species	Untreated	Treated[a]	LC_{50} ng/cm^2	Relative susceptibility[b]
Trichoplusia ni	0.0	65.6±7.3	1792.2	1.00
Heliothis zea	0.7±0.3	18.1±1.8	6103.9	3.40
Spodoptera exigua	0.0	20.8±3.6	5194.8	2.90
Pseudoplusia includens	0.0	88.8±3.9	1649.4	0.92
Anticarsia gemmatalis	0.7±0.6	78.7±2.4	1428.6	0.80
Plathypena scabra	3.0±1.0	75.4±5.5[c]	1.1	0.01

[a] Larvae exposed to 2077.9 ng/cm^2 of diet except where noted.
[b] Based upon LC_{50} of *T. ni*.
[c] Larvae exposed to 10 ng/cm^2.

dens). Thus, weight gains of larvae exposed on treated diets provided another indication that *H. zea* and *S. exigua* were less susceptible and that *P. scabra, P. includens*, and *A. gemmatalis* were more susceptible than *T. ni* to *B. thuringiensis*. Species that were less susceptible than *T. ni* (*H. zea, S. exigua*), as indicated by lower relative mortality (Tables 2, 3), gained considerably more weight on treated diets than the more susceptible species (*P. scabra, P. includens, A. gemmatalis*), whether they were exposed as 1-, or 4-day-old larvae. The approximate gains in weight over the initial weight for larvae exposed on treated diet as 1-day-old larvae were: *P. scabra*, 1×; *A. gemmatalis*, 10×; *P. includens*, 20×; *T. ni*, 30×; *S. exigua*, 300×; *H. zea*, 325×. The approximate weight gains for 4-day-old larvae were: *T. ni*, 2×; *H. zea* 11×; *S. exigua*, 25×. Significant weight gains were not obtained for 4-day-old larvae of the most sensitive species (*A. gemmatalis, P. includens*, and *P. scabra*). Therefore, the ranking of relative susceptibility of the caterpillar species to *T. ni* based upon weights of larvae exposed on treated diets (Table 4) was identical to the ranking based on mortality (Tables 2, 3): *P. scabra, A. gemmatalis, P. includens, T. ni, S. exigua*, and *H. zea*.

DISCUSSION.—In a previous (unpublished) study conducted in 1967 in which we compared the activity of *B. thuringiensis* var. *thuringiensis* formulated as Thuricide® 90TS against neonatal larvae of *H. zea, S. exigua, H. virescens*, and *T. ni*, we found that *H. zea* and *S. exigua* were ca. 4.3× and 2.9× more resistant than *T. ni* and that *H. virescens* was ca. 3× more susceptible than *T. ni*. Later Rogoff et al. (1969) evaluated the insecticidal activity of 31 strains of *B. thuringiensis* against *T. ni, H. zea, Galleria mellonella* (L.), *Culex pipiens* L., and *Musca domestica* L. However, this was a qualitative screen, and direct quantitative comparisons of relative susceptibility could not be made from their data. Dulmage (1970) reported a quantitative comparative study of an isolate (HD-1-R-11) of *B. thuringiensis* var. *alesti* (= *kurstaki*) assayed against *Pectinophora gossypiella* (Saunders), *H. virescens*, and *H. zea* in which he used *T. ni* as the standard. His data concerning the relative susceptibility of *H. zea* to *T. ni* are in close agreement with ours, namely, *H. zea* was ca. 3.0× less susceptible than *T. ni*. (Larvae of *P. gossypiella* and *H. virescens* were ca. 15× and 36× more susceptible than *T. ni*.)

From the results of our laboratory evaluation of relative susceptibility, *P. scabra* should be controlled easily on soybeans at the rates normally used to control *T. ni* (½–1 lb/acre, the equivalent of 3.6–7.3 billion IU/acre). Also, lower rates (ca. ¼–¾/acre, 1.8–5.4 billion IU/acre) may be equally as effective, especially if the treatment is directed at neonatal or 1st-intar *P. scabra*. Larvae of *A. gemmatalis* and *P. includens* also should be controlled at the rates normally used to control *T. ni*, and, as with *P. scabra*, effectiveness would be enhanced by treating early instars. Results of recent field studies conducted on soybeans by Yearian et al. (1973) support our laboratory find-

Table 4.—Average weight (±S$\bar{X}$) of larvae that survived continuous exposure to *Bacillus thuringiensis*-treated diet or to untreated diet.[a]

	Weight of 1-day-old larvae[b]		Weight of 4-day-old larvae[c]	
Species	Treated diet	Untreated diet	Treated diet	Untreated diet
Trichoplusia ni	19.7±4.4	219.0±20.8	18.1±2.9	301.8±6.2
Heliothis zea	96.2±7.2	605.0±32.1	108.1±36.9	551.2±28.8
Spodoptera exigua	98.1±14.1	257.6±18.4	249.1±15.4	206.9±19.2
Anticarsia gemmatalis	3.3±2.1	23.5±5.0	8.3±0.7	204.4±14.1
Pseudoplusia includens	6.2±1.8	40.8±7.3	8.7±0.6	235.3±17.8
Plathypena scabra[d]	0.4±0.1	8.9±2.6	7.3±3.7	57.9±14.0

[a] Initial larval weight of 1-day-old larvae ranged from 0.2–0.5 mg; that of 4-day-old larvae ranged from 9–12 mg/larva.
[b] One-day-old larvae exposed to dose of 90.9 ng/cm^2 were weighed after 10 days of exposure on diet.
[c] Four-day-old larvae exposed to dose of 2077.9 ng/cm^2 diet were weighed after 7 days of exposure on diet.
[d] Exposed to 0.8 ng/cm^2.

ings: in the tests they conducted in Arkansas, *P. scabra* were reduced an average 76% (range 58–93%) using *B. thuringiensis* var. *kurstaki* at a rate of 1–4 billion IU/acre, and these same rates reduced *A. gemmatalis* and *P. includens* an average of 65 or 69%, respectively. On the other hand, larvae of *S. exigua* or *H. zea* probably cannot be controlled on soybeans at rates normally used to control *T. ni*. From our laboratory results, rates of 3–4 lb/acre, the insecticidal equivalent of 22–30 billion IU/acre, would be needed.

Our current information indicates that control of practically all caterpillar pests of soybeans might be obtained using 2 safe, selective insecticides; namely, *B. thuringiensis* for control of defoliators and the *Heliothis* nucleopolyhedrosis virus, *Baculovirus heliothis* (Ignoffo et al. 1976) for control of pod-feeding *Heliothis* spp. The virus has been registered recently for use against *Heliothis* species on cotton, and our preliminary field tests indicate that it could be effectively used to control *Heliothis* species on soybeans.

ACKNOWLEDGMENT.—We acknowledge the valuable contribution of W. A. Dickerson and his insectary staff of the Biological Control of Insects Research Unit, who supplied the insects used in these studies.

REFERENCES CITED

Dulmage, H. T. 1970. Insecticidal activity of HD-1, a new isolate of *Bacillus thuringiensis* var. *alesti*. J. Invertebr. Pathol. 15: 232–9.

Dulmage, H. T., and R. A. Rhodes. 1971. Production of pathogens in artificial media. p. 507–40. *In* H. D. Burges and N. W. Hussey [Ed.] Microbial Control of Insects and Mites, Academic Press, London-New York. 861 pp.

Dulmage, H. T., O. P. Boening, C. S. Rehnborg, and G. D. Hansen. 1971. A proposed standardized bioassay for formulations of *Bacillus thuringiensis* based on the international unit. J. Invertebr. Pathol. 18: 240–5.

Greene, G. L., W. C. Leppla, and W. A. Dickerson. 1976. Velvetbean caterpillar: a rearing procedure and artificial medium. J. Econ. Entomol. (In press.)

Ignoffo, C. M. 1966. Insect viruses. Pages 501–30 *in* C. N. Smith [Ed.] Insect Colonization and Mass Production, Academic Press, New York. 618 pp.

Ignoffo, C. M., and O. P. Boening. 1970. Compartmented disposable plastic trays for rearing insects. J. Econ. Entomol. 63: 1696–7.

Ignoffo, C. M., C. Garcia, and V. A. Gasparotto. 1968. Sensitivity of larvae of the cabbage looper, *Trichoplusia ni*, to *Bacillus thuringiensis*. J. Invertebr. Pathol. 11: 97–103.

Ignoffo, C. M., N. L. Marston, B. Puttler, D. L. Hostetter, G. D. Thomas, and W. A. Dickerson. 1976. Natural biotic agents controlling insect pests of Missouri soybeans. Proc. World Soybean Res. Conf., Urbana, Ill., 1975. (In press.)

Libby, J. L., and R. K. Chapman. 1971. *Bacillus thuringiensis* strains for cabbage caterpillar control in Wisconsin. J. Invertebr. Pathol. 18: 114–6.

Rogoff, M. H., C. M. Ignoffo, S. Singer, I. Gard, and A. P. Prieto. 1969. Insecticidal activity of thirty-one strains of *Bacillus* against five insect species. Ibid. 14: 122–9.

Sommerville, H. J., Y. Tanada, and E. M. Omi. 1970. Lethal effect of purified spore and crystalline endotoxin preparations of *Bacillus thuringiensis* on several lepidopterous insects. Ibid. 16: 241–8.

van den Bosch, R. 1972. The costs of poisons. Environ. 14: 18–31.

Yearian, W. C., J. M. Livingston, and S. Y. Young. 1973. Field evaluation of *Bacillus thuringiensis* for control of selected lepidopterous foliage feeders on soybeans. Arkansas Agric. Exp. Stn. Rep. Ser. 212. 8 pp.